

WHITE PAPER

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# Outsmarting **Silent Data Corruption** in AI Processors With Two-Stage Detection

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## 1 | What Is Silent Data Corruption?

Silent Data Corruption (SDC) is an industry challenge plaguing data centers worldwide. This phenomenon is caused by untraceable hardware failures, hence the term “silent,” making it notoriously hard to identify. SDCs don’t trigger exception mechanisms or leave any record in system logs. Their resulting corrupted data can keep propagating while they stay undetected, causing cascading failures that often require substantial efforts to root cause.

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**SDC rate is much higher than software engineers expect, undermining the hardware reliability they used to take for granted.**

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Recent publications by hyperscalers such as Google<sup>1</sup> and Meta<sup>2</sup> shed some light on the extent of the problem. They report that approximately one in a thousand machines in their fleets is affected by SDC. While they only provide estimates, Alibaba<sup>3</sup> recently published exact statistics revealing 361 Defective Parts Per Million (DPPM) that caused SDCs in their cloud system. All in all, SDC rate is much higher than software engineers expect, undermining the hardware reliability they used to take for granted.

As infrastructure rapidly shifts towards artificial intelligence (AI), these numbers are sure to rise. AI systems rely heavily on large volumes of data and complex computations to function effectively. SDC can introduce subtle errors into the data or computations without immediate detection, leading to incorrect outputs and potentially faulty decision-making. As AI systems are increasingly integrated into critical applications, the impact and magnitude of undetected errors can be catastrophic. The complexity and opacity of AI models further intensify the issue, making it challenging to trace and correct the source of corruption.

Within AI systems, an SDC can create parameter corruption, where AI model parameters are corrupted and their original values are altered. Therefore, ensuring data integrity and implementing robust fault detection mechanisms are essential to maintaining the reliability and trustworthiness of AI systems.

Unlike some other types of data errors, SDC occurs primarily due to compute faults rather than memory bitflips. Redundancy-based methods, such as Error Correcting Code (ECC), can protect against corruption in stored and communicated data. However, these techniques cannot detect SDCs, which are typically caused by erroneous execution of processor instructions.

IEEE’s<sup>4</sup> deeper inspection of the cause of the problem revealed that “Timing errors, design bugs, and manufacturing defects occur inside a microprocessor chip and may cause silent errors.” Alibaba<sup>3</sup> provided further details, stating that “SDCs exhibit a substantial prevalence in particular workloads, exposing five vulnerable features, namely arithmetic logic computation, vector operations, floating point calculation, cache coherency and transactional memory.”

## 2 | SDC Is on the Rise, Escaping Detection Despite Extensive Testing

The problem has gotten worse in recent years as new process nodes and complex architectures have pushed the boundaries of semiconductor technology, outpacing silicon testing capabilities. Conventional Scan, ATPG, BIST and functional testing are often unable to detect the causes of silent data corruption, making it an inevitable fault, however destructive. These tests were sufficient for detecting distinct defects, providing simple pass/fail results. Nonetheless, they are not sensitive enough to identify some subtle SDC-related anomalies, which stem from semiconductor process variations.

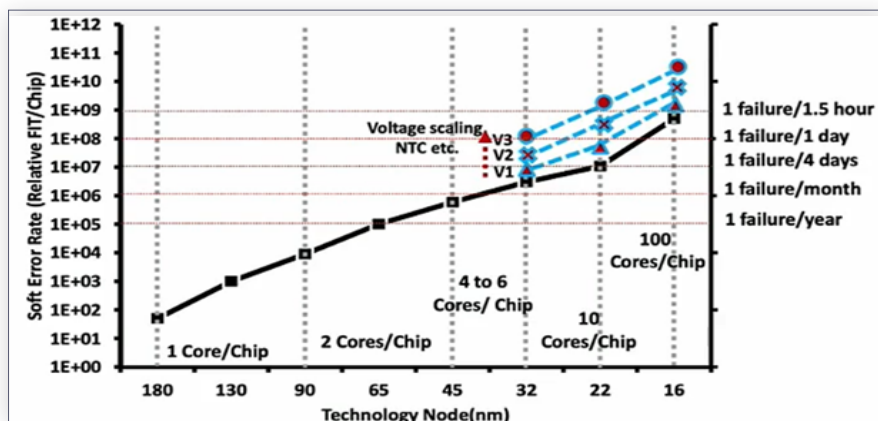


Figure 1: Soft errors such as SDC have increased from 1 failure per year @ 65nm to 1 failure every 1.5 hours @ 16nm<sup>5</sup>

Such silicon testing shortcomings result in defects escaping detection time and again, underscoring the necessity of in-field monitoring. Unlike the chip manufacturing stage, testing during fleet operation factors in silicon aging and insufficient guard bands exposed by real workloads. Using today's best-known methods, testing is performed either in situ or through periodic maintenance, yet both of which still have significant blind spots:

- If performed with canary circuits, in-situ testing is blind to real, critical path timing margins, which might decrease due to aging and process variations. This consideration has become crucial with the increase in on-chip variation within a device, as mentioned in the “MRHIEP.”<sup>6</sup>
- Periodic maintenance testing might not be sensitive enough, mostly identifying distinct failures while overlooking subtler SDC-related issues. It also lacks the real-life conditions that characterize in-situ monitoring, as tested devices are removed from the fleet.

Another reason for the rise in SDC rates is the unprecedented scale of compute-intensive workloads, such as AI applications, ML model training, and inference computation. These tasks often run at peak CPU performance, increasing the probability of silent data corruption. The following table outlines the potential causes of SDC during two stages of a chip's lifespan—manufacturing and in-field operation.

**Another reason for the rise in SDC rates is the unprecedented scale of compute-intensive workloads, such as AI applications, ML model training, and inference computation.**

| Chip Manufacturing                         | Fleet Operation             |
|--------------------------------------------|-----------------------------|
| Process nodes with high transistor density | Silicon aging               |
| Semiconductor process variations           | Insufficient margins        |
| Silicon testing shortcomings               | Large-scale infrastructures |
| Latent defects                             | Compute-intensive workloads |

Potential causes of SDC during two stages of a chip’s lifespan

### 3 | A Silent Killer: Why SDC is Such a Big Problem

Why should anyone worry about 361 DPPM or even several thousands of them? At a low scale, such numbers are considered normal. However, in fleets that can have millions of servers, SDC occurrences are frequent enough to threaten the integrity of vital services. For example, if a generative AI data system runs on thousands of devices simultaneously, a single error in one of them can lead to many system-level failures.

In addition to direct revenue losses caused by SDC, hyperscalers are especially concerned about its silent nature, which makes it unpredictable. With users that expect zero downtime and flawless operation 24/7, operators cannot afford this uncertainty. As a result, they are on a quest for a remedy that can help them regain confidence in their fleets. As Intel<sup>7</sup> stated in a recent article, “In a data center running multiple millions of servers, 24 hours a day, the rare event becomes an expected occurrence, and the system administrator must be proactive in managing the event”.

One attempt to bypass the problem relies on redundant computations that can compensate for any data corruption. In this approach, the same functionality is executed on multiple cores, and only if they all produce the same result it is considered correct. While this method can prevent SDC propagation, it is extremely expensive as it requires an impractical amount of hardware at large scales.

**Many SDC investigations end inconclusively despite substantial investment, effectively perpetuating the uncertainty.**

Another reason SDC is such a big problem is the innate complexity of its debugging process, which can take months to complete, as Meta<sup>8</sup> reported. Troubleshooting a fault that leaves no trace requires ingenuity, often alongside extensive resources. To make matters worse, many SDC investigations end inconclusively despite substantial investment, effectively perpetuating the uncertainty. A recent publication by Broadcom<sup>9</sup> showed that as many as 50% of their SDC investigations ended with “No Trouble Found.”

## 4 | The Consequences of Unchecked SDCs

Unchecked SDCs can lead to a variety of unpredictable faults, from barely noticeable miscalculations to malfunctions with severe business impact. Here are a few real-life examples:

- **Miscalculated mathematical operations in a defective CPU led to database file loss - Meta<sup>8</sup>**  
 File decompression wasn't performed as the defective CPU erroneously calculated its size, producing zero instead of 156 when executing `Int(1.153)`.
- **A storage application reporting checksum mismatch of user data – Alibaba Cloud<sup>3</sup>**  
 A defective processor instruction related to checksum calculation gave wrong results intermittently.
- **Defective cache coherence producing wrong checksums – Alibaba Cloud<sup>3</sup>**  
 A client thread shared some data and its checksum with a daemon thread, which sometimes got inconsistent information due to a cache coherency issue.
- **Incorrect hashing calculation in a faulty processor – Alibaba Cloud<sup>3</sup>**  
 The investigation led to a hash map that manages the program's metadata, helping to pinpoint the hashing calculation defect.
- **Manufacturing defects exposed by rare instructions – Google<sup>1</sup>**  
 An innocuous change to a low-level library caused servers to make heavier use of otherwise rarely executed instructions, which ran incorrectly due to manufacturing defects.

Another consequence of SDC is the extraordinary effort required to troubleshoot it. As stated by Google<sup>1</sup> – “The problem is serious enough for us to have applied many engineer-decades to it.” Meta<sup>8</sup> has documented a detailed example of its SDC debugging process, revealing a time-consuming effort by multiple engineering teams. They had to go all the way from a high-level Scala workload to low-level assembly code through nearly ten different stages. Unlike C/C++, Scala is not compiled directly into machine code, forcing a complex flow from Scala to Java to Java Byte Code (JBC) to assembly. This nontrivial translation required several tools:

- Scala compiler to convert Scala into Java code
- Hotspot profiling library to print JIT-compiled code
- GCJ compiler to convert JBC into assembly (deprecated since 2008)

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Meta also needed to perform multiple modifications while reverse engineering the assembly code to create a stable reproducer, for example:

- Eliminate absolute address references
- Observe compiler optimizations
- Control unintended branches
- Handle memory offsets
- Manage stack frames

This troubleshooting effort by Meta was impressive, but it cannot be the way forward. It is unscalable as it requires too much manual work from expert engineers over a long time. Moreover, it has no power to prevent the problem, only to debug it after it has already manifested.

As stated in the “MRHIEP”<sup>6</sup>, “The industry sentiment is that we are only seeing the tip of the iceberg of this fault type, and new techniques will be required over the next five to ten years to drive down their rate of occurrence.”

## 5 | What if We Had Multi-Stage Deep Data Detection

There is another way, which involves multi-stage detection during both chip manufacturing and in-field operation. It allows chipmakers to reclaim product reliability while fleet operators regain confidence in their hardware by significantly reducing the SDC rate. Monitoring multiple stages with deep data parametric visibility can compensate for the elusive nature of the problem, improving the overall probability of detection.

For that, we need higher granularity during silicon testing, providing parametric grading that considers process variations and predicted performance margins rather than mere pass/fail results. Such a method flags outlier devices susceptible to SDC even if they pass all conventional tests, preventing ‘walking-wounded’ systems to reach the market.

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We also need continuous monitoring in the field, enabling real-time intervention when design margins fall dangerously low. Chips equipped with this technology can automatically adjust voltage or frequency to compensate for aging and environmental effects, match workload fluctuations, and prevent SDC.

## 6 | Outsmarting SDC With Two-Stage Detection

A multi-stage approach that combines continuous on-chip monitoring with deep data analytics is well aligned with the conclusions of industry leaders. In the words of Google<sup>1</sup>, for example, “For detecting mercurial cores as quickly as possible; in effect, testing becomes part of the full lifecycle of a CPU, not just an issue for vendors or burn-in testing.”

proteanTecs has introduced multi-stage monitoring, offering SDC prevention solutions for different stages of a chip’s lifespan:

- proteanTecs Outlier Detection™ application for tests during chip manufacturing
- proteanTecs RTHM™ (Real-Time Health Monitoring) for monitoring during fleet operation

Outlier Detection can greatly reduce SDC-related errors when testing the silicon at ATE, System-Level Test or System Test, while RTHM helps prevent SDC in the field.

The company’s technology delivers an industry-only approach that combines parametric on-chip monitoring at the physical layer of the silicon, with advanced and continuous analytics, connecting the HW and SW layers for the first time.

To learn more, download the *proteanTecs On-Chip Monitoring and Deep Data Analytics System White Paper* [here](#).

The solution operates in both test and mission modes, providing visibility of performance, quality, and reliability under functional workloads. This is especially needed in AI applications, where diverse and scaling workloads drive surging challenges in data integrity.

## 7 | proteanTecs Outlier Detection™ - Preventing SDC With ML-Powered Silicon Testing

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**It learns the normal behavior by processing agent readings using ML, and identifies anomalies undetectable by other methods.**

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proteanTecs Outlier Detection is based on parametric measurements of on-chip agents, which monitor multiple parameters, including the timing of millions of IC paths. Edge software that runs on the tester provides a real-time, inline assessment, with analytics and insights provided on a cloud-based platform. The application learns the normal behavior by processing agent readings using ML, and identifies anomalies undetectable by conventional methods, during semiconductor testing at ATE. The solution also employs time series analysis at system-level test, to ensure quality in long-running tasks.

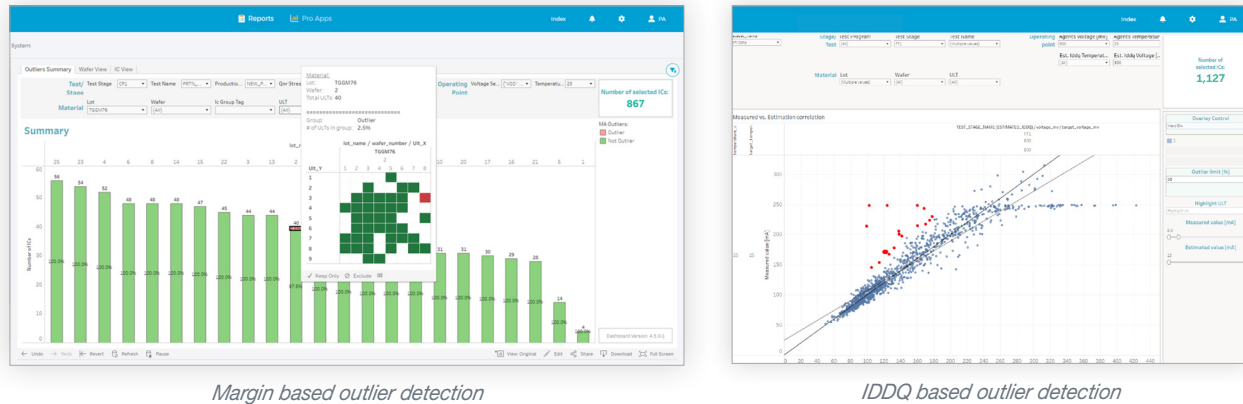


Figure 2: Outliers detected at ATE, preventing latent defects in the field

With proteanTecs, a personalized and high-sensitivity ML model provides granular expected results for IDDQ and timing margins for each device (expected vs. actual). This adds a predictive dimension to quality assurance, without compromising yield. The technique goes far beyond pass/fail testing, and is able to detect latent defect that go unnoticed today, especially in low-margin paths.

At system-level test, proteanTecs employs time-series analysis for detecting SDC, which often does not manifest in shorter tests due to its inconsistent nature. In the case of SDC, the allotted cycle time is insufficient to conclude the logic calculations under functional workloads. Using proteanTecs, which measures in real-time the delta between clock cycle to the worst-case logic path, timing margins are parametrically measured.

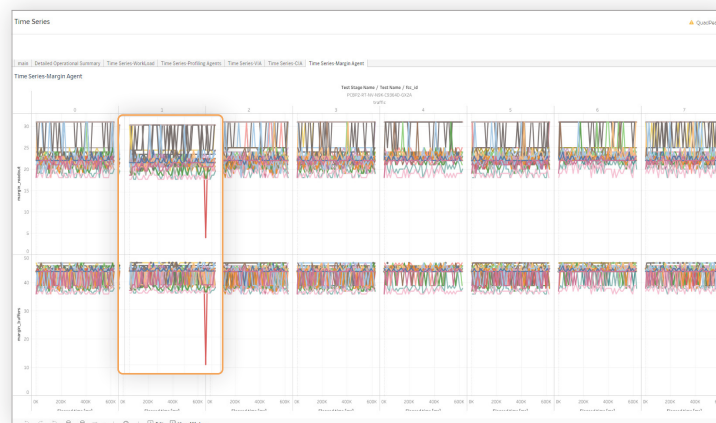


Figure 3: proteanTecs Outlier Detection identifies severe margin drops in a device that is likely to cause SDC; Time series analysis at system-level test

In one case, the solution detected DUTs with severe margin drops from 22 time units to only 4, leading the customer to discard them. These SDC-prone devices had passed all conventional tests as their performance was marginal rather than unacceptable. However, after some aging in the field, they were likely to run out of guard-band and cause SDC. Further analysis revealed that proteanTecs Outlier Detection helped to reduce the potential SDC rate by up to 40% for the customer.

Read the customer case study [here](#).

## 8 | proteanTecs Real-Time Health Monitoring™ - Fleet Monitoring With Real-Time SDC Prevention

Common SDC prevention methods typically rely on periodic maintenance, which has a costly overhead as it tests all systems regardless of their health. However, even fleet operators who are willing to pay the price of excessive testing are not immune. Unfortunately, they still encounter cases of SDC, which is often detected only after the faults manifest in the field, during lifetime operation. This can be caused by accelerated aging, application stress, environmental effects, or still undetected latent defects.

proteanTecs RTHM takes a different approach, providing Predictive and Prescriptive Maintenance instead of preventive maintenance. RTHM is a FW application that identifies issues in real-time, with recommendations for corrective actions and real-time failure prevention interrupts. The detected events are not actual errors yet, but they might accumulate to a low chip health index, which often precedes SDCs. This solution uses dedicated thresholds to deduce when margins get dangerously low, as depicted below.

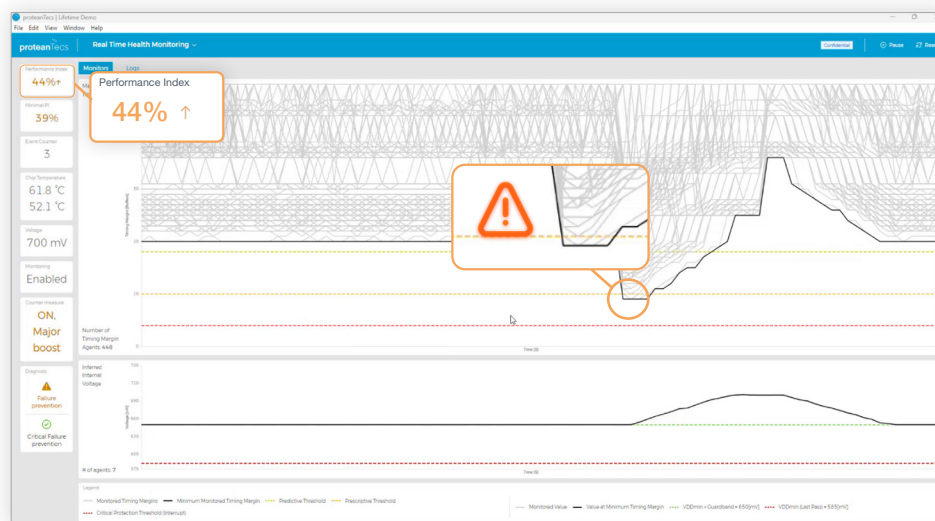


Figure 4: *proteanTecs RTHM real-time indication of a severe margin drop in a device that might cause SDC, during functional operation in the field (this is a FW application, the visualization provided above is from a real system)*

RTHM provides a Performance Index (PI) by processing agent readings alongside other inputs using advanced real-time algorithms. The PI is continuously monitored and reflects the proximity to failure, factoring in:

- Actual workloads
- Aging
- Affected areas
- Clock and/or power domains
- Previous events

**A low health index will trigger an interrupt, allowing the BMC to decide whether to take corrective action.**

When the failure is imminent, an interrupt signal is set, allowing the application to decide whether to take corrective action given the current system status, and also log the location of where the imminent failure is about to occur. In some configurations, RTHM can take corrective action based on low PI levels, even before the imminent failure.

To learn more, download the RTHM white paper [here](#).

## 9 | Conclusion

Silent data corruption is on the rise following advancements in semiconductor technology. The explosion in artificial intelligence for speech, image, video and text processing leads to a growing complexity and diversity of hardware systems, bringing an increased risk to data integrity. Undetected manufacturing defects, accelerated aging, or environmental factors can lead to data corruption, during storage, transmission, or processing, and result in unintended changes in information.

Now more than ever, SDC-related faults translate to revenue loss as well as uncertainty, which fleet operators cannot afford. Traditional approaches to prevent SDC during silicon manufacturing and data center operation fail to provide adequate reliability. The proteanTecs two-stage detection approach offers a novel solution with ML-powered Outlier Detection for semiconductor validation and unique Real-Time Health Monitoring for in-field testing. This technology can significantly reduce the SDC rate, ensuring greater hardware reliability in AI workload-intensive applications.

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